Nathaniel Esubonteng

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EDUCATION

Stanford University **Stanford, CA**

B.S. in Mathematics and Computer Science *Expected Jun 2028*

Clubs**: National Society of Black Engineers, Akwaaba, Quant Gym, Stanford Association for Computing Machinery, BlackGen Capital, ColorStack**

Skills: Python, C++, Java, JavaScript/TypeScript, SQL; React, Flask, Node.js/Express; AWS (S3, Bedrock), Docker, Linux, Git, REST APIs; LLM and RAG systems, NumPy, Pandas

EXPERIENCE

CareerVillage.org

Software Engineering Intern – Coach AI Platform *June 2026 - Present*

- Engineer and ship backend features for Coach, CareerVillage's AI-powered career coaching platform, developing APIs, agent tool-calling functionality, automated tests, and production fixes
- Optimize local development across the Coach and CareerVillage repositories using Docker Compose and Infisical, resolving cross-service networking conflicts and reducing setup friction for engineers
- Build backend integrations and routing logic for WhatsApp and RCS, expanding Coach's AI career-guidance capabilities beyond its existing web experience

Rutgers University – Zhu Research Group **Newark, NJ**

Research Intern *June 2026 - Present*

- Extending Anvil, an active-learning framework for machine-learning interatomic potentials, by implementing a plugin interface and completing a partially-built state machine that selects candidate structures for DFT calculations and orchestrates ensemble model training
- Collaborate via a PR-based workflow on a shared research codebase to close the active-learning loop end-to-end ahead of a planned methods paper
- Built Python utilities (Pandas, NumPy) to batch-rename files, validate simulation runs, and aggregate large CSV datasets, ensuring data integrity and reproducibility for graduate-level physics simulation

Jane Street **New York City, NY**

FOCUS Fellow Participant *April 2026 - May 2026*

- Selected for Jane Street's fully funded FOCUS program at its New York City headquarters.
- Participated in professional-development programming, office hours, and networking sessions with employees across trading, software engineering, and strategy and product.
- Gained exposure to Jane Street's quantitative problem-solving culture and career pathways in finance and technology.

Amazon Inc. **Newark, NJ**

Audible Software Engineering Intern – Voder (Virtual Voice Studio) Team *Feb 2025 – Sep 2025*

- Optimized AWS cloud infrastructure by developing high-throughput Python CLI tools for S3 and Bedrock API automation, reducing operational latency by 4x and eliminating manual bottlenecks
- Brainstormed and built command line interfaces for S3 upload/download and Bedrock API calls, allowing teammates to execute cloud operations up to 4x faster without relying on the AWS console
- Reduced time-to-market for AI-driven products by 22% through the integration of automated data streams into the VVS narration pipeline, enabling rapid iterative backtesting of model outputs.

Audible Data Capture Intern – DRAGON Team *Aug 2024 – Feb 2025*

- Designed a high-efficiency data visualization interface for the Celyx console, streamlining the analysis of 15M+ daily clickstream events to identify anomalous data patterns.
- Developed structured SQL-style query mockups and filtering logic to demonstrate improved cross-checking workflows, cutting analysis time by ~50% and reducing formatting inconsistencies by ~70%
- Collaborated with software engineers and product managers on Feature FEAT-9761, refining usability and presentation details that earned 90%+ positive feedback at the Product & Insights Summit

The Gem Project **Newark, NJ**

Youth Civic Engagement Fellow *Jan 2023 – Dec 2024*

- Led Newark's campaign to lower the school board voting age to 16, mobilizing 7,000+ students and gaining recognition from The New York Times and MSNBC for advancing youth-led democratic reform.

PROJECTS

LUMINARY — Best Use of WebSpatial, SCU Hack for Humanity 2026 (79 projects, 500 participants)

React, TypeScript, Python, Flask, Node.js, Manim, ElevenLabs, Gemini, WebSpatial SDK *Mar 2026*

- Built a full-stack AR classroom for Apple Vision Pro with a conversational AI teacher, live Manim math animations rendered from natural-language prompts, and synchronized ElevenLabs voice narration
- Engineered a TypeScript/Express Manim rendering service that generates and executes Python animation code on demand, proxied through a Flask backend with MD5-keyed file caching for lessons, audio, and video
- Architected a 3-service distributed system (React frontend, Flask API, Node render service) with BroadcastChannel-synchronized multi-window spatial scenes for visionOS via WebSpatial SDK